

Exp. No. 4.2

Design and Simulation of 4:1 Multiplexer using Verilog code in Xilinx

Aim

To design and simulate a 4:1 multiplexer using Verilog HDL and verify its functionality using the Xilinx Vivado simulator.

Verilog Code

```
module mux_4to1(
    input [1:0] sel,
    input d0,
    input d1,
    input d2,
    input d3,
    output reg y
);
    always @(*)
    begin
        case(sel)
            2'b00: y = d0;
            2'b01: y = d1;
            2'b10: y = d2;
            2'b11: y = d3;
            default: y = 1'b0;
        endcase
    end
endmodule
```

Testbench

```
`timescale 1ns / 1ps
module mux_4to1_tb;
    reg [1:0] sel;
    reg d0, d1, d2, d3;
    wire y;

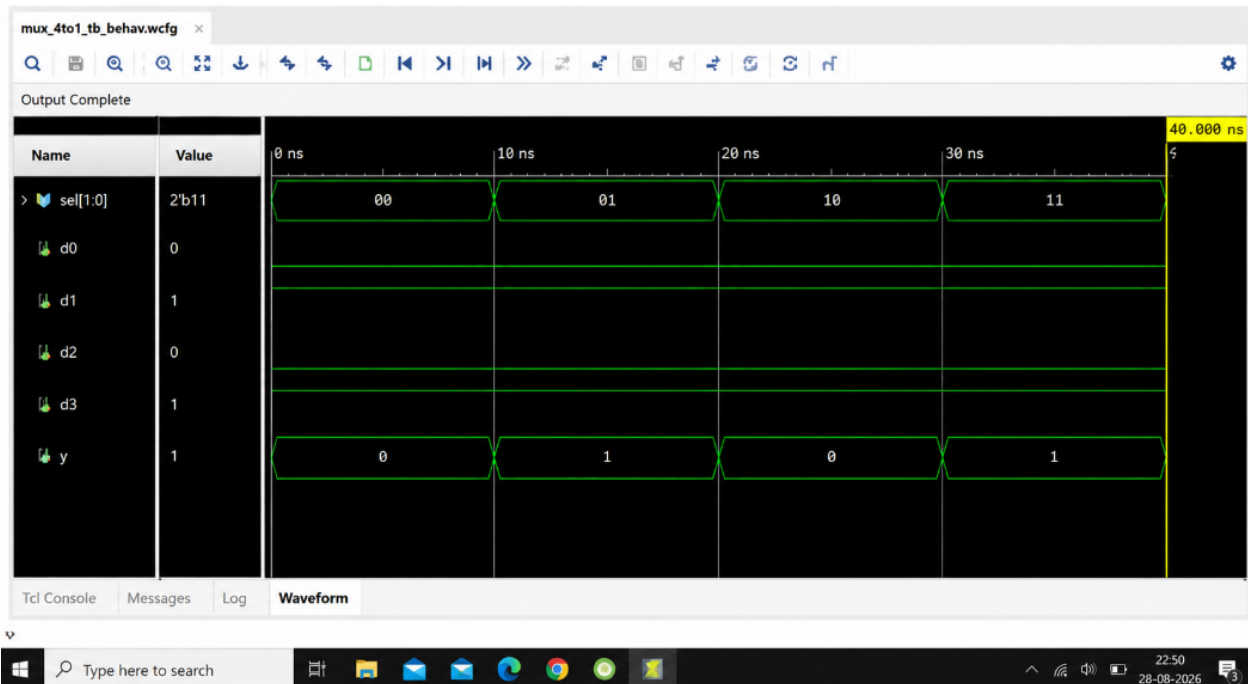
    mux_4to1 uut (
        .sel(sel),
        .d0(d0),
        .d1(d1),
        .d2(d2),
        .d3(d3),
        .y(y)
    );

    initial begin
        d0 = 0;
        d1 = 1;
        d2 = 0;
        d3 = 1;

        sel = 2'b00; #10;
        sel = 2'b01; #10;
        sel = 2'b10; #10;
        sel = 2'b11; #10;
        $finish;
    end
endmodule
```

```
end  
endmodule
```

Output



Result: The 4:1 multiplexer was designed and simulated successfully in Xilinx Vivado. With d0=0, d1=1, d2=0, d3=1, for sel=00,01,10,11 the outputs obtained were y=0,1,0,1 respectively, matching the expected truth table.